

ZERO-TRUST OFFLINE ENCRYPTED EXAM SYSTEM

An Open Architecture for Leak-Resistant Digital Examinations
A Defense-in-Depth Security Framework

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ZERO-TRUST OFFLINE ENCRYPTED EXAM SYSTEM

An Open Architecture for Leak-Resistant Digital Examinations

A Defense-in-Depth Security Framework Supporting Open-Source Hardware and Software Implementations

◆ 1. OBJECTIVE

To eliminate question paper leaks by combining Zero-Trust Security, Offline Cryptography, Secure Hardware, Human Expert-Curated Question Banks, Cryptographically Verifiable Random Selection, and Complete Auditability.

This architecture is designed to ensure that no individual, server, administrator, examination center, or device can independently compromise the confidentiality of examination papers.

◆ 2. HUMAN EXPERT-CURATED QUESTION BANK

- Subject Experts, Professors, Teachers, and Lecturers from across the country independently submit confidential examination questions.
- The Examination Authority reviews, validates, and approves all submitted questions.
- Duplicate, invalid, ambiguous, outdated, or low-quality questions are removed through expert review and automated similarity detection.
- Every approved question is assigned a permanent Question ID (e.g., Q0001–Q1000).
- Each approved question is securely tagged with:
 - Subject
 - Topic
 - Difficulty Level
 - Marks
 - Language Version
 - Question Version

- AI-assisted similarity detection, combined with expert review, identifies and removes duplicate or substantially repeated questions before the Question Bank is finalized.
- Once finalized, the Question Bank is digitally signed, cryptographically locked, and securely stored for the cryptographically verifiable random selection process.

◆ 3. SECURE RANDOM QUESTION SELECTION

- The system does NOT generate questions using AI.
- Instead, an automated cryptographically secure selection engine randomly selects questions only from the approved Question Bank.
- Selection uses a Cryptographically Secure Random Number Generator (CSPRNG).
- The selection process follows predefined examination rules such as:
 - Topic Coverage
 - Difficulty Balance
 - Marks Distribution
 - Syllabus Coverage

- Example:

Question Bank

Q0001 – Q1000

↓

Random Selection

↓

Q0003

Q0011

Q0125

Q0150

↓

Final Set A

The same process independently creates Sets B, C, and D.

- Every selection event is digitally signed and recorded in immutable audit logs in accordance with the Examination Authority's retention policy.

◆ 4. ENCRYPTED EXAM PACKAGE

- Final Sets A, B, C, and D are converted into digitally signed and encrypted examination packages.
- Every examination package is verified using cryptographic hash validation and digital signature verification.
- The encrypted examination packages are securely prepared for authenticated provisioning to authorized examination devices through the Trusted Provisioning Infrastructure.
- Until the officially scheduled examination time, every examination package remains completely encrypted and unreadable.

◆ 5. SECURE PROVISIONING INFRASTRUCTURE

- Examination Authorities securely operate one or more Trusted Provisioning Stations (TPS) inside physically protected and access-controlled facilities.
- The approved Question Bank remains securely maintained, digitally signed, and cryptographically protected within the Trusted Provisioning Station.
- Final encrypted examination packages (Sets A/B/C/D) are generated from the approved Question Bank within a predefined secure operational window before the official examination start time. Every examination package is digitally signed and cryptographically verified prior to distribution.
- Final encrypted examination packages (Sets A/B/C/D) are securely provisioned to all authorized examination devices through authenticated and encrypted communication channels defined by the Secure Network Architecture.
- Every examination device independently verifies the digital signature, cryptographic integrity, and authenticity of each received examination package before accepting it.
- Trusted Provisioning Stations maintain complete cryptographically signed audit logs for every provisioning event, including package version, device identity, timestamp, verification status, and authorized administrative actions.
- Following successful provisioning verification for all authorized examination devices, every examination device automatically enters Hardware-Enforced Offline Examination Mode. From this point onward, examination devices cannot communicate directly with one another or perform any unauthorized external communication during the examination.

- Examination packages, provisioning records, and audit evidence remain securely preserved to support post-examination verification, authorized review, forensic analysis, dispute resolution, and other official Examination Authority functions in accordance with the Examination Authority's approved retention policy.
- Any unauthorized access attempt is cryptographically logged together with the device identity, timestamp, authentication result, and, where permitted by applicable law and organizational policy, optional security evidence such as camera images.
- After the examination has officially concluded, authorized administrative synchronization may be performed only through authenticated procedures defined by the Secure Network Architecture for answer synchronization, audit log collection, software or firmware updates, and preparation for future examination cycles. No examination package may be modified, replaced, or reissued during this administrative synchronization process.

6. CORE HARDWARE SECURITY

- Dedicated, reusable E-Ink examination devices are designed exclusively for secure examination purposes.
- Paperless operation with a low eye-strain E-Ink display provides improved readability while minimizing visual fatigue.
- Secure Boot prevents unauthorized operating systems, firmware, or software from loading during device startup.
- A Hardware Root of Trust (e.g., TPM or Secure Element) verifies device identity, firmware integrity, and cryptographic credentials before the device becomes operational.
- Only digitally signed and authenticated firmware is permitted to execute on the device.
- Physical tamper detection automatically locks the device and records a cryptographically signed security event if unauthorized hardware manipulation is detected.
- Hardware security mechanisms are designed to resist firmware modification, unauthorized debugging, secure key extraction, and other forms of physical compromise.
- Student examination devices provide only a USB Type-C charging interface. Data transfer through the USB interface is permanently disabled at both the hardware and firmware levels.
- Trusted Provisioning Stations may include additional administrative interfaces (such as secure USB ports, smart-card readers, or Hardware Security Module interfaces) exclusively for authorized administrative operations. These interfaces remain physically protected and accessible only to authorized personnel.

◆ 7. BATTERY MANAGEMENT

- Every examination device shall provide a minimum battery capacity sufficient for at least 15–20 hours of continuous examination operation and approximately 5–7 days of standby availability under normal operating conditions.
- Every examination center shall maintain certified spare examination devices together with fully charged Smart Charging Docks to ensure uninterrupted examination continuity.
- Smart Charging Docks automatically recharge devices while performing battery health verification, device diagnostics, firmware version verification, device integrity checks, and operational readiness assessments before every examination.
- Wireless charging technologies may be supported where operationally practical; however, wired USB Type-C charging remains the recommended primary charging method for reliability, simplicity, and ease of maintenance.

◆ 8. SECURE NETWORK ARCHITECTURE

- Student examination devices never establish direct communication with the public Internet.
- Student examination devices communicate only with authorized Trusted Provisioning Stations through mutually authenticated and encrypted communication channels.
- Trusted Provisioning Stations act as the only authorized communication gateway between examination devices and the Government's secure examination infrastructure.
- Trusted Provisioning Stations may connect to Government secure servers through dedicated secure networks, Virtual Private Networks (VPNs), or other authenticated encrypted communication channels approved by the Examination Authority.
- All communications between Trusted Provisioning Stations and Government secure servers are digitally authenticated, end-to-end encrypted, and cryptographically logged.
- Following successful provisioning and verification, every student examination device automatically enters Hardware-Enforced Offline Examination Mode, during which all external communication interfaces remain disabled until authorized administrative synchronization is initiated after the examination.
- Post-examination answer synchronization, audit log submission, software updates, and future examination provisioning may occur only through authenticated administrative operations initiated by the Trusted Provisioning Station. Student examination devices never independently access external networks.
- All network activities, provisioning events, synchronization operations, firmware updates, and administrative actions are permanently recorded within cryptographically signed audit logs.

◆ 9. OFFLINE CRYPTOGRAPHIC ACCESS CONTROL

- Examination papers remain encrypted inside every examination device.
- Offline Cryptographic Time-Lock automatically unlocks the authorized examination package at the officially scheduled examination time.
- Internet connectivity is not required for examination package decryption.
- Examination devices continue functioning normally even during temporary or prolonged network outages.
- After secure provisioning, every examination device automatically enters Hardware-Enforced Offline Examination Mode. During the examination, all external communication interfaces remain disabled. Network connectivity may be restored only through authenticated administrative procedures defined by the Secure Network Architecture.

◆ 10. STUDENT AUTHENTICATION

- Every examination device is assigned a unique and permanent Hardware Device ID.
- Each candidate is securely bound to the assigned examination device through the candidate's Admit Card QR Code (or an equivalent secure assignment mechanism approved by the Examination Authority).
- Multi-factor identity verification, including 3D Face and/or Iris Authentication with Infrared Liveness Detection, verifies the presence of the authorized candidate before examination access is granted.
- Presentation attacks, including photographs, videos, printed images, masks, or other spoofing attempts, are detected and rejected.
- Device assignment, candidate authentication, examination center, seat number, timestamp, authentication outcome, and other relevant security events are securely recorded within cryptographically protected audit logs.

◆ 11. OFFLINE SUPERVISOR AUTHORIZATION

- Three consecutive authentication failures automatically lock the examination device.
- No universal Master Key or universal bypass mechanism exists.
- Device recovery requires an authorized Supervisor to successfully complete multi-factor administrative authentication using:

- Supervisor Biometric Authentication
- Offline Security Smart Card or Hardware Security Token
- The examination device locally verifies the supervisor's credentials without requiring Internet connectivity and, upon successful verification, authorizes a controlled recovery procedure or an additional candidate authentication attempt in accordance with the Examination Authority's security policy.
- During the examination, supervisor authorization shall not permit access to examination content. It only enables controlled administrative recovery or candidate re-authentication.
- After the examination has concluded, examination content may be made available only to authorized personnel, such as examiners, evaluators, or designated administrators, through authenticated and role-based access controls for answer evaluation, audit, dispute resolution, or other officially authorized examination processes.
- All supervisor authorization events, recovery procedures, post-examination access events, and administrative actions are cryptographically logged with the user identity, device identity, timestamp, authentication outcome, and access purpose.
- Internet connectivity is not required for supervisor authorization.

12. EXAM CENTER DEVICE FAILURE RECOVERY

- Every examination center maintains certified spare examination devices that are securely provisioned and ready for authorized deployment.
- If an examination device becomes damaged, defective, or otherwise unusable during the examination:
 - The candidate's identity is reverified.
 - The Admit Card QR Code (or an equivalent secure assignment mechanism) is rescanned.
 - The candidate is securely reassigned to a certified spare examination device through the authorized recovery procedure.
 - The assigned examination package is restored and made available only after successful candidate authentication and verification.
- Where applicable, previously recorded candidate responses are securely recovered or synchronized in accordance with the Examination Authority's approved examination procedures.
- All recovery operations, device reassignments, authentication events, and administrative actions are cryptographically logged for audit and forensic purposes.

- Examination continuity is preserved without exposing examination content or compromising examination integrity.

◆ 13. ZERO-TRUST CRYPTOGRAPHIC SECURITY

- AES-256 encryption protects examination packages and sensitive data at rest and during authorized transmission.
- Public-Key Cryptography provides secure key exchange, digital signatures, and authentication.
- Split-Key Authorization ensures that no single individual, device, or server can independently decrypt or authorize access to examination packages.
- Hardware Security Modules (HSMs) or equivalent hardware-backed secure key storage protect master cryptographic material.
- Remote Attestation verifies the integrity and authenticity of authorized devices before examination provisioning or access.
- Cryptographic Hash Verification detects unauthorized modification of examination packages, firmware, configuration data, and other security-critical components.
- All cryptographic operations are performed using industry-standard, publicly reviewed cryptographic algorithms and security protocols.

◆ 14. ANTI-LEAK PROTECTION

- Dynamic Multi-Set Fallback (Sets A/B/C/D) provides operational resilience against suspected examination compromise.
- If an examination compromise is suspected before the official examination begins, the Examination Authority may authorize activation of an alternative pre-provisioned encrypted examination package in accordance with approved security procedures.
- Personalized anonymous dynamic watermarking uniquely identifies each displayed examination instance for forensic tracing without publicly revealing the candidate's identity.
- Cryptographically signed and tamper-evident audit logs preserve a complete record of security events, administrative actions, and examination activities to support authorized forensic investigation, integrity verification, and dispute resolution.

◆ 15. ANTI-CHEATING FEATURES

- AI-assisted camera and reflection detection may identify suspected unauthorized recording attempts or suspicious optical devices and promptly alert invigilators for manual verification.
- AI-assisted detection serves only as a supplementary security layer and does not independently determine examination misconduct or replace human supervision, cryptographic protections, or other core security controls.

◆ 16. POST-EXAM SECURITY

- Temporary decryption keys used exclusively for examination access are permanently destroyed immediately after the examination concludes.
- Temporary decrypted memory used during examination processing is securely erased to prevent unauthorized recovery of transient data.
- The authorized examination package accessed during the examination (e.g., Set A, if selected) remains preserved in its authenticated post-examination state to support answer evaluation, audit, forensic investigation, dispute resolution, and other officially authorized review processes. It is not automatically re-encrypted.
- Any unused examination packages (e.g., Sets B/C/D) remain securely encrypted and cryptographically protected unless officially authorized for rescheduled examinations, audit, investigation, legal proceedings, or other approved Examination Authority purposes.
- Immutable cryptographically signed audit logs remain available for authorized forensic investigation, examination integrity verification, and accountability.
- Examination records, audit evidence, and associated security logs remain preserved until their retention period expires under the Examination Authority's officially approved retention policy, after which they may be securely archived or securely deleted in accordance with applicable law and organizational policy.

◆ 17. SECURITY PRINCIPLES

- ✓ Zero Trust Architecture
- ✓ Defense in Depth
- ✓ Offline-by-Default Security
- ✓ Cryptographic Verification

- ✓ Human Expert-Curated Question Bank
- ✓ Cryptographically Secure Random Question Selection
- ✓ Secure Hardware Root of Trust
- ✓ Role-Based Access Control
- ✓ Complete Auditability
- ✓ Privacy Protection
- ✓ Open Architecture
- ✓ Open Knowledge and Collaborative Improvement

◆ 18. OPEN KNOWLEDGE & LICENSING DECLARATION

- This architecture is made available under the Creative Commons Attribution–ShareAlike 4.0 International (CC BY-SA 4.0) license to encourage global collaboration, implementation, improvement, and knowledge sharing.
- It is intended to support open-source hardware and software implementations and may be used, studied, implemented, modified, translated, and redistributed by governments, universities, educational institutions, cybersecurity researchers, developers, and other organizations worldwide, subject to the terms of the CC BY-SA 4.0 license.
- Security should rely on transparency, cryptography, independent verification, and Zero-Trust principles—not on secrecy.

◆ 19. EXAM RECORD PRESERVATION & ADULT CONTINUITY

- No examination package or official examination record is automatically deleted after the examination.
- Once the officially scheduled examination time unlocks a specific examination package (e.g., Set A, if selected), that package remains preserved in its authenticated post-examination state for answer evaluation, audit, forensic investigation, dispute resolution, and other officially authorized review processes. It is never automatically re-encrypted.
- Any unused examination packages (e.g., Sets B/C/D) remain securely encrypted and cryptographically protected unless officially authorized for rescheduled examinations, audit, investigation, legal proceedings, or other approved Examination Authority purposes.

- Examination records, candidate responses, device logs, authentication logs, audit records, and other official examination evidence shall be retained for the period determined by the Examination Authority in accordance with applicable law and organizational policy.
- Before any authorized device reset or re-provisioning for a future examination cycle, and only after the examination has officially concluded, the Examination Authority may securely back up and transfer the required examination records from the examination device to its authorized secure servers or cloud infrastructure through authenticated and encrypted communication channels. No examination records or examination packages shall be transferred before the official conclusion of the examination.
- Following successful archival or verification, an authorized secure reset may permanently remove the previous examination data from the device, allowing it to be securely re-provisioned and reused for future examinations.
- This preservation framework ensures complete traceability, supports post-examination investigations, protects candidates' rights, enables independent verification of every examination session, and maintains the integrity of the official examination record.

20. OPEN LICENSE

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This architecture is intended to remain a permanent open knowledge resource for humanity. Its purpose is to encourage global collaboration in building secure, transparent, privacy-preserving, and leak-resistant examination systems for the benefit of present and future generations.

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